

ISLAMIC UNIVERSITY OF TECHNOLOGY

BoardBazar, Gazipur, Dhaka, 1704, Bangladesh

CSE 4410- Lab Project Proposal

Department of Computer Science And Engineering

BSc. in Software Engineering

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Title of the project:

PharmaTrack

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Brief Overview of the Project:

PharmaTrack is a comprehensive solution designed to streamline the tracking and management of pharmaceutical products, suppliers, purchases, sales, and customer information. PharmaTrack aims to automate inventory updates, alert on low stock levels of critical medications, and provide robust reporting functionalities to facilitate informed decision-making within pharmaceutical organizations. It aims to address the unique challenges faced by pharmaceutical organizations in managing their inventory effectively, ensuring the availability of critical medications, and maintaining regulatory compliance.

Features:

- **Product Management:** Enable users to add, update, and delete pharmaceutical products with detailed information including name, description, dosage, expiry date, batch number, and storage requirements.
- **Inventory Tracking:** Monitor real-time quantity available and location for each pharmaceutical product, ensuring accurate stock levels and efficient inventory management.
- **Supplier Management:** Manage essential supplier information including name, contact details, certifications, and product catalogs to facilitate streamlined procurement processes.
- **Purchase Orders:** Record purchase orders for pharmaceutical products, capturing supplier details, purchase date, quantity, batch numbers, and purchase prices to ensure traceability and regulatory compliance.
- **Sales Orders:** Capture sales transactions of pharmaceutical products including product sold, sale date, quantity, patient information (if applicable), and sale price for accurate revenue tracking and inventory depletion.
- **Inventory Alerts:** Receive real-time alerts when inventory levels of critical medications fall below predefined thresholds, enabling timely replenishment and preventing stockouts.
- **Reporting and Analytics:** Generate comprehensive reports on various aspects such as inventory value, product sales, expiration tracking, low stock items, and purchase/sales trends to support strategic decision-making and regulatory compliance.
- **User Management:** Administer user roles and permissions tailored to the pharmaceutical industry's regulatory requirements, ensuring data security, access control, and compliance with privacy regulations such as HIPAA (Health Insurance Portability and Accountability Act).
- **Data Integrity:** Implement robust data validation and integrity checks to maintain accurate and reliable data, adhering to regulatory standards and best practices in pharmaceutical inventory management.

Tools and Technology:

- **Database Management System:** Oracle will be used for data storage and management.
- **Programming Languages:** SQL will be used for database querying and manipulation and optionally, a backend programming language such as Java will be used for application logic. HTML and CSS will be used for user interface.
- **Version Control:** Git will be used for version control to track changes and collaborate effectively.

ERD Diagram:

